Finding the Fault in a Proof

- A proof step is **circular** when its reason is the thing being proved, or something that depends on it. A step is **unsupported** when its reason is not a given, a definition, a postulate, or an earlier line — "it looks that way in the picture" is not a reason.
- For each item: say *which line* fails and *why*, then repair it. Two of the items also ask you to redo the arithmetic correctly.
- Diagrams show labelling only; all measurements are in the problem text.

1. Two lines cross at V. A student writes the proof below.

Statement	**Reason**
1. $\overleftrightarrow{AC}$ and $\overleftrightarrow{BD}$ intersect at V	1. Given
2. $\angle AVB$ and $\angle CVD$ are a pair of opposite angles	2. Definition of vertical angles
3. $\angle AVB \cong \angle CVD$	3. Because $\angle AVB$ and $\angle CVD$ are congruent

(a) Name the line that fails, and say what is wrong with its reason.

(b) Write a reason that would make the proof valid.

2. Ari is asked to solve $4x - 6 = 2x + 10$ and writes:

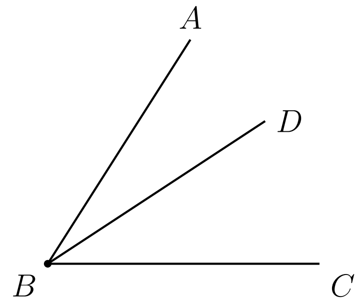
Statement	**Reason**
1. $4x - 6 = 2x + 10$	1. Given
2. $4x = 2x + 10$	2. Addition property of equality
3. $2x = 10$	3. Subtraction property of equality
4. $x = 5$	4. Division property of equality

(a) Ari's answer is $x = 5$. Name the line where the mistake happens and say what rule was broken.

(b) Redo the work correctly and give the true value of x.

Answer: ________________

3. $\overrightarrow{BD}$ is drawn inside $\angle ABC$. A student writes the proof below. The only given is that $\overrightarrow{BD}$ lies in the interior of $\angle ABC$.



Statement	Reason
1. $\overrightarrow{BD}$ is interior to $\angle ABC$	1. Given
2. $\angle ABD \cong \angle DBC$	2. The two halves look equal in the figure
3. $m\angle ABD = m\angle DBC$	3. Definition of congruent angles

(a) Which line is unsupported, and why does its reason not count?

(b) State what would have to be added to the given information to make the proof valid, and name the reason line 2 would then use.

4. M is the midpoint of $\overline{PQ}$, with $PM = 5x - 3$ and $MQ = 2x + 9$ in centimetres. Sam writes:

Statement	**Reason**
1. M is the midpoint of $\overline{PQ}$	1. Given
2. $5x - 3 = 2x + 9$	2. Definition of midpoint
3. $3x = 6$	3. Subtraction property of equality
4. $x = 2$	4. Division property of equality

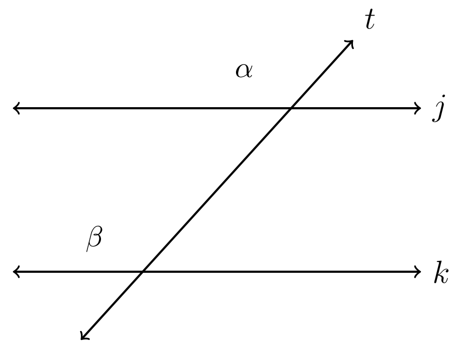
(a) Lines 1 and 2 are correct. Name the first faulty line and describe the mistake exactly.

(b) Redo the work correctly and give the true value of x.

Answer: ________________

5. A student is asked why two lines j and k cut by a transversal are parallel, and argues:

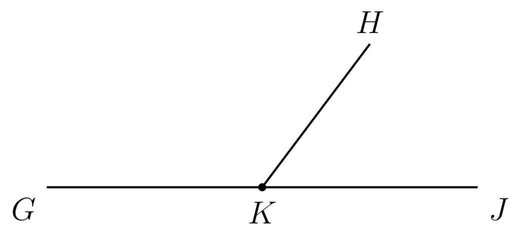
> "The corresponding angles are congruent, so $j \parallel k$. And I know the corresponding angles are congruent because j and k are parallel."



(a) Explain why this argument proves nothing.

(b) Rewrite it as a valid argument. Say what you are given at the start and which statement you are entitled to conclude.

6. $\angle GKH$ and $\angle HKJ$ form a linear pair. In degrees, $m\angle GKH = 3x+20$ and $m\angle HKJ = 5x$. A student writes:



Statement	Reason
1. $\angle GKH$ and $\angle HKJ$ form a linear pair	1. Given
2. $m\angle GKH = m\angle HKJ$	2. Vertical angles are congruent
3. $3x + 20 = 5x$, so $x = 10$	3. Properties of equality

(a) Line 2 is unsupported. Say why the reason given does not apply here, and state the correct relationship and the postulate that supplies it.

(b) Using the correct relationship, find the true value of x.

Answer: __________________